

Spectral Probing of the GigaPath Backbone

Jacobian Singular Value Analysis as a Dynamical Systems Diagnostic

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1 Motivation: Layers as a Dynamical System

We regard the depth-wise application of transformer blocks in the GigaPath backbone as a discrete-time dynamical system. Let $\mathbf{h}_l \in \mathbb{R}^{N \times d}$ denote the token-sequence hidden state at layer l , where N is the number of tokens (including the CLS token) and $d = 1536$ is the embedding dimension. Each transformer block defines the recurrence

$$\mathbf{h}_{l+1} = F_l(\mathbf{h}_l; \theta_l) = \mathbf{h}_l + G_l(\mathbf{h}_l; \theta_l), \quad (1)$$

where G_l is the residual branch (self-attention $\rightarrow$ layer-scale $\rightarrow$ MLP $\rightarrow$ layer-scale) and θ_l collects the parameters of block l . The total depth is $L = 40$.

Attractor hypothesis. After self-supervised pre-training the system (1) has converged to a regime in which input representations are mapped to a low-dimensional attractor. The geometry of this attractor is characterised by the *Lyapunov spectrum*: the rates at which infinitesimally nearby trajectories (i.e. nearby inputs) diverge or converge as they propagate through the layers.

2 GigaPath Backbone Architecture

The GigaPath tile encoder is a DINOv2-giant Vision Transformer (ViT-g) with $L = 40$ layers, embedding dimension $d = 1536$, 24 attention heads (head dimension $d_h = 64$), and a SwiGLU MLP with hidden dimension 4096 (gated via an 8192-wide intermediate). The full architecture is shown in Figure 1.

3 Which Jacobian?

Two natural Jacobians can be defined at every layer:

Symbol	Definition	Measures
$\mathbf{J}_l^{(h)} = \frac{\partial \mathbf{h}_{l+1}}{\partial \mathbf{h}_l}$	State-space Jacobian	Sensitivity of output to <i>hidden state</i> perturbations
$\mathbf{J}_l^{(\theta)} = \frac{\partial \mathbf{h}_{l+1}}{\partial \theta_l}$	Parameter Jacobian	Sensitivity of output to <i>weight</i> perturbations

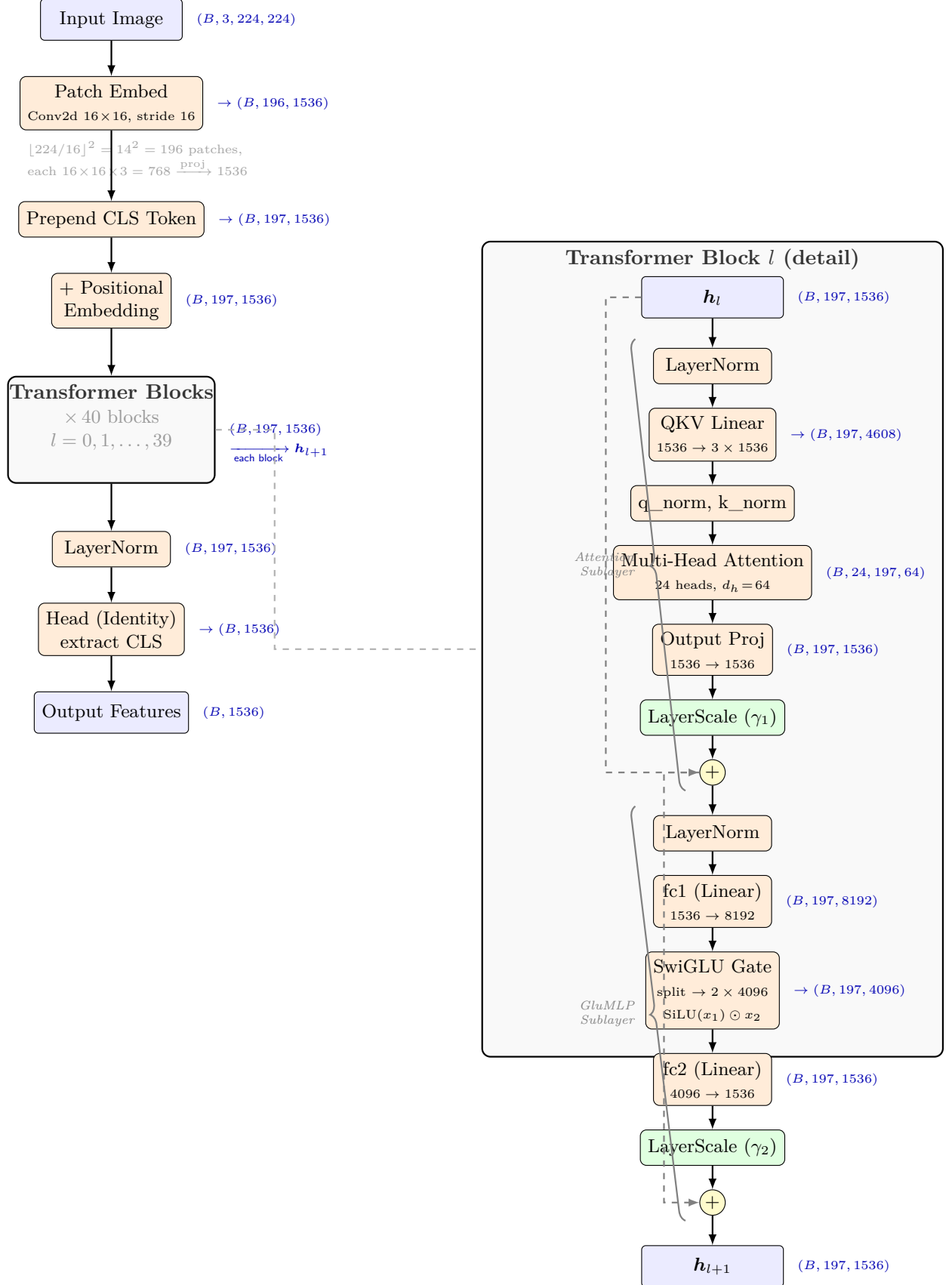


Figure 1: Architecture of the GigaPath tile backbone (DINOv2-giant ViT-g/14, $L = 40$, $d = 1536$, 24 heads). **Left:** full forward pipeline from input image to output feature vector. **Right:** expanded view of a single transformer block l , showing the attention sublayer with QKV $\rightarrow$ MHSA $\rightarrow$ Proj $\rightarrow$ LayerScale $\rightarrow$ Residual, and the GluMLP sublayer with fc1 $\rightarrow$ SwiGLU $\rightarrow$ fc2 $\rightarrow$ LayerScale $\rightarrow$ Residual. Dashed arrows indicate residual skip connections. All tensor shapes are annotated in

For Lyapunov analysis, use the state-space Jacobian $\mathbf{J}_l^{(h)}$. Lyapunov exponents characterise how nearby *trajectories* (hidden states produced by nearby inputs) diverge or converge. This is entirely governed by $\mathbf{J}_l^{(h)}$. The parameter Jacobian $\mathbf{J}_l^{(\theta)}$ measures Fisher-information-type sensitivities, which is a different (though related) diagnostic.

The residual structure (1) implies

$$\mathbf{J}_l^{(h)} = \mathbf{I} + \frac{\partial G_l}{\partial \mathbf{h}_l}, \quad (2)$$

so the Jacobian is a perturbation of the identity.

4 Singular Values and the Lyapunov Spectrum

4.1 Notation

Let $\sigma_i^{(l)}$ denote the i -th singular value of $\mathbf{J}_l^{(h)}$, i.e.

$$\sigma_i^{(l)} = \sigma_i(\mathbf{J}_l^{(h)}) = \sqrt{\lambda_i(\mathbf{J}_l^{(h)\top} \mathbf{J}_l^{(h)})}, \quad (3)$$

where $\lambda_i(\cdot)$ denotes the i -th eigenvalue (in decreasing order). In other words, $\sigma_i^{(l)}$ is the i -th singular value of the layer- l Jacobian, measuring the local expansion ($\sigma > 1$) or contraction ($\sigma < 1$) rate in the i -th principal direction at that layer.

- $\sigma_i^{(l)} > 1$: the i -th direction is **expanded** by layer l — information is amplified.
- $\sigma_i^{(l)} < 1$: the i -th direction is **contracted** — information is discarded.
- $\sigma_i^{(l)} \approx 1$: the i -th direction is approximately **preserved** (dynamical isometry).

4.2 Finite-depth Lyapunov exponents

The (finite-depth) Lyapunov exponents of the system are defined by the singular values of the composed mapping $\mathbf{J}^{[0:L]} = \prod_{l=0}^{L-1} \mathbf{J}_l^{(h)}$. Concretely, by the multiplicative ergodic theorem (Oseledets, 1968), the Lyapunov exponents are

$$\Lambda_i = \lim_{L \rightarrow \infty} \frac{1}{2L} \log \lambda_i \left(\prod_{l=0}^{L-1} \mathbf{J}_l^{(h)\top} \mathbf{J}_l^{(h)} \right). \quad (4)$$

For a finite-depth network, we replace the limit with a finite sum and define the *empirical Lyapunov exponents* as

$$\hat{\Lambda}_i = \frac{1}{L} \sum_{l=0}^{L-1} \log \sigma_i^{(l)}. \quad (5)$$

This is an approximation; strictly it assumes the singular-value directions align across layers, but it provides a useful summary statistic.

Key observables.

Observable	Interpretation
$\sigma_{\max}^{(l)} \gg 1$	Layer l strongly amplifies certain directions
$\sigma_{\min}^{(l)} \ll 1$	Layer l strongly contracts certain directions
$\kappa^{(l)} = \sigma_{\max}^{(l)} / \sigma_{\min}^{(l)} \gg 1$	High condition number; anisotropic geometry
All $\sigma_i^{(l)} \approx 1$	Dynamical isometry (ideal for signal propagation)
$\hat{\Lambda}_i > 0$	Net expansion over the full depth — chaos-like regime
$\hat{\Lambda}_i < 0$	Net contraction — attractor-like regime
$\hat{\Lambda}_i \approx 0$	Critical regime (edge of chaos)

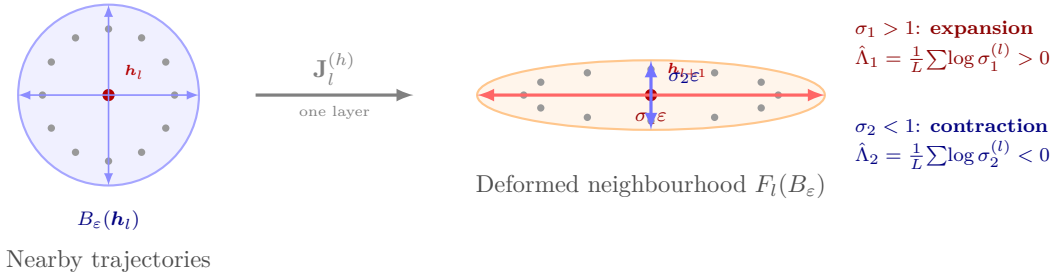


Figure 2: Geometric meaning of Lyapunov exponents in 2-D. The Jacobian $\mathbf{J}_l^{(h)}$ maps a ball of nearby hidden states (left) to an ellipse (right). The semi-axes are $\sigma_1 \varepsilon$ and $\sigma_2 \varepsilon$, where σ_1, σ_2 are the singular values. Directions with $\sigma > 1$ (red, positive Lyapunov exponent) are *expanded* — nearby trajectories diverge. Directions with $\sigma < 1$ (blue, negative Lyapunov exponent) are *contracted* — nearby trajectories converge. After L layers the net expansion/contraction is $\exp(L\hat{\Lambda}_i)$, where $\hat{\Lambda}_i = \frac{1}{L} \sum_l \log \sigma_i^{(l)}$.

In three or more dimensions, an additional possibility arises: some directions may be *neutral* ($\sigma \approx 1$) while others expand or contract. Figure 3 shows the 3-D analogue: a unit sphere is mapped to an ellipsoid with one expanding, one neutral, and one contracting axis.

Composition of Jacobians and axis misalignment. When two layers are composed, the singular values of $\mathbf{J}_2 \mathbf{J}_1$ are *not* simply the products of the individual singular values unless the principal axes of $\mathbf{J}_1$ and $\mathbf{J}_2$ are aligned. In general, the right singular vectors of $\mathbf{J}_1$ (the directions it stretches) must be projected onto the left singular vectors of $\mathbf{J}_2$ (the directions $\mathbf{J}_2$ acts on). Because attention layers rotate the representation at each step, the axes are typically misaligned, and the composed map acquires a different stability structure from either factor.

Figure 4 demonstrates this in 3-D. Each individual Jacobian ($\mathbf{J}_1$ and $\mathbf{J}_2$) has *one unstable, one neutral, and one stable* direction. But the composed map $\mathbf{J}_2 \circ \mathbf{J}_1$ has *one unstable, zero neutral, and two stable* directions: the neutral direction of $\mathbf{J}_1$ is rotated by $\mathbf{J}_2$ so that it partially overlaps with a contracting direction, causing it to become stable in the composition. More generally, axis misalignment tends to *increase the number of stable directions*, which is precisely the mechanism by which deep networks converge to low-dimensional attractors.

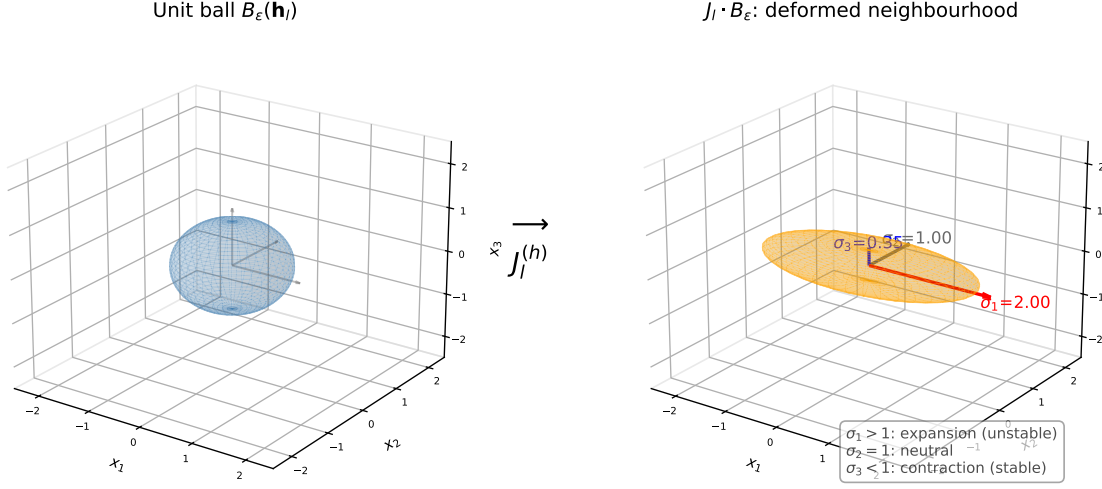


Figure 3: 3-D Lyapunov deformation. A unit sphere of nearby hidden states is mapped by the Jacobian $\mathbf{J}_l^{(h)}$ to an ellipsoid with semi-axes $\sigma_1 = 2.0$ (red, expansion), $\sigma_2 = 1.0$ (gray, neutral), and $\sigma_3 = 0.35$ (blue, contraction). In the 1536-dimensional ViT hidden space, each layer produces 1536 such singular values, defining the instantaneous local geometry of the dynamics.

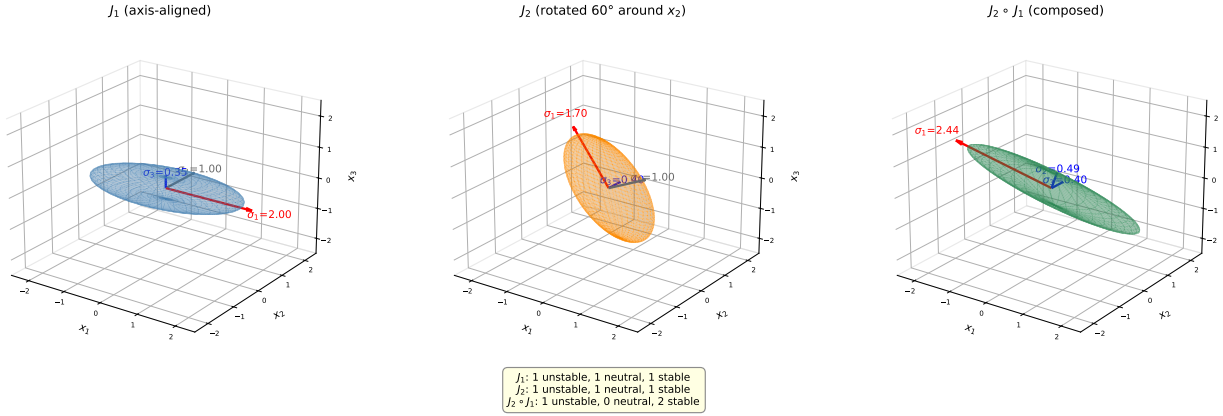


Figure 4: Composition of two Jacobians with misaligned principal axes. **Left:** $\mathbf{J}_1 = \text{diag}(2.0, 1.0, 0.35)$ (axis-aligned; 1 unstable, 1 neutral, 1 stable). **Centre:** $\mathbf{J}_2$ has singular values $(1.7, 1.0, 0.4)$ but its axes are rotated 60 around x_2 (1 unstable, 1 neutral, 1 stable). **Right:** the composed map $\mathbf{J}_2 \circ \mathbf{J}_1$ has singular values $(2.44, 0.49, 0.40)$ — *one unstable, zero neutral, two stable*. The neutral direction vanishes because $\mathbf{J}_1$'s neutral axis is rotated into a contracting direction of $\mathbf{J}_2$. This illustrates how depth and axis misalignment progressively increase contractivity.

4.3 Why the residual structure matters

From (2), $\mathbf{J}_l^{(h)} = \mathbf{I} + \partial G_l / \partial \mathbf{h}_l$, so $\sigma_i^{(l)} = 1 + \delta_i^{(l)}$ where $\delta_i^{(l)}$ are the singular-value perturbations introduced by the residual branch. The interesting diagnostic is the *deviation spectrum* $\{\delta_i^{(l)}\}$, or equivalently $\{\log \sigma_i^{(l)}\}$, plotted as a function of layer index l .

4.4 Input dependence

Because transformers use attention, $\mathbf{J}_l^{(h)}$ depends on the input $\mathbf{h}_l$, which itself depends on the data point. The spectral probe therefore produces a *per-sample* local Lyapunov spectrum. In practice, we aggregate by averaging singular values (or their logs) over a mini-batch.

5 Feasibility of the Attractor Hypothesis

The hypothesis is well-supported by prior work:

1. **Deep Information Propagation** (Schoenholz et al., 2017): at initialisation, deep networks must operate at the edge of chaos (all Lyapunov exponents ≈ 0) for trainability.
2. **Dynamical isometry** (Saxe et al., 2014; Pennington et al., 2018): well-trained networks have Jacobian singular values clustered near 1.
3. **Residual networks** (Yang & Schoenholz, 2018): residual connections bias $\mathbf{J}$ toward the identity, naturally promoting dynamical isometry.
4. **For a frozen pre-trained ViT** such as GigaPath, the expectation is that most $\sigma_i^{(l)} \approx 1$, with a few extreme values encoding the dominant feature-transformation modes. The largest singular values correspond to the directions the network *amplifies* (semantically important features), and the smallest to the directions it *discards*.

A thorough mathematical treatment of this perspective is given in Boris Hanin’s lecture series at the University of Chicago (Lecture 1), where the transport of data through the layers of a deep network is formalised as a dynamical system whose Lyapunov exponents govern the geometry of the latent-space embedding.

Even in low-dimensional settings, the coexistence of expanding and contracting directions produces remarkably complex geometry. The Lorenz system (Figure 5) is a 3-D ODE with one positive and two negative Lyapunov exponents: nearby trajectories diverge along the unstable manifold while being compressed toward the attractor, producing the iconic “butterfly” — a fractal object confined to a thin, folded sheet in phase space. In a 1536-dimensional ViT hidden space, the interplay of hundreds of expanding and contracting singular-value directions could give rise to analogously intricate structure.

6 Efficient Computation

6.1 Dimensionality

With $N = 197$ tokens (196 patches + CLS) and $d = 1536$, the full Jacobian $\mathbf{J}_l^{(h)} \in \mathbb{R}^{Nd \times Nd}$ has dimension $302,592 \times 302,592$. Materialising or decomposing this matrix is infeasible.

Lorenz Attractor ($\sigma=10$, $\rho=28$, $\beta=8/3$)

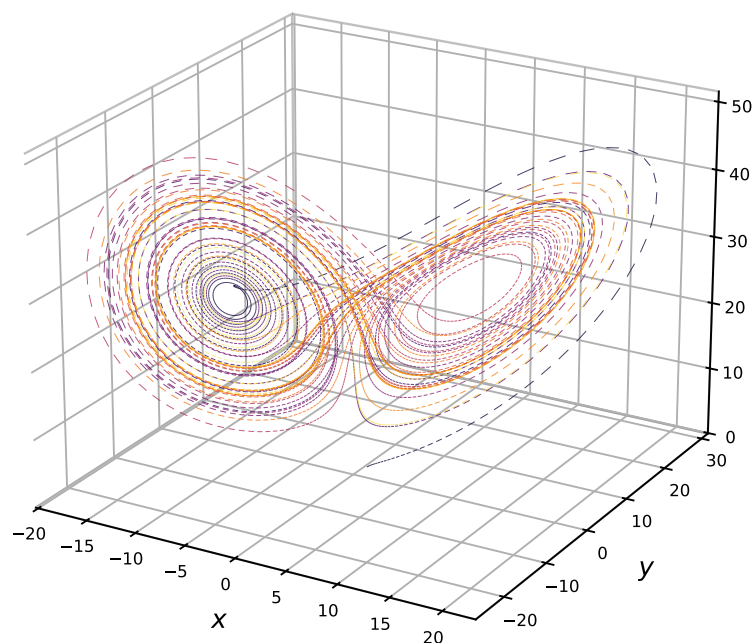


Figure 5: The Lorenz attractor: a canonical example of deterministic chaos in three dimensions. One positive Lyapunov exponent ($\Lambda_1 > 0$, expansion) combined with two negative exponents ($\Lambda_2 < 0$, $\Lambda_3 < 0$, contraction) produces a strange attractor — trajectories are stretched and folded onto a fractal set. This illustrates how even a few competing expansion/contraction directions can generate highly complex dynamics. The spectral probe of the GigaPath backbone measures the analogous singular-value structure in $d = 1536$ dimensions.

However, restricting to the CLS token is both tractable and informative, because self-attention allows the CLS token to read from *every* patch token at every layer. Figure 6 illustrates this information flow.

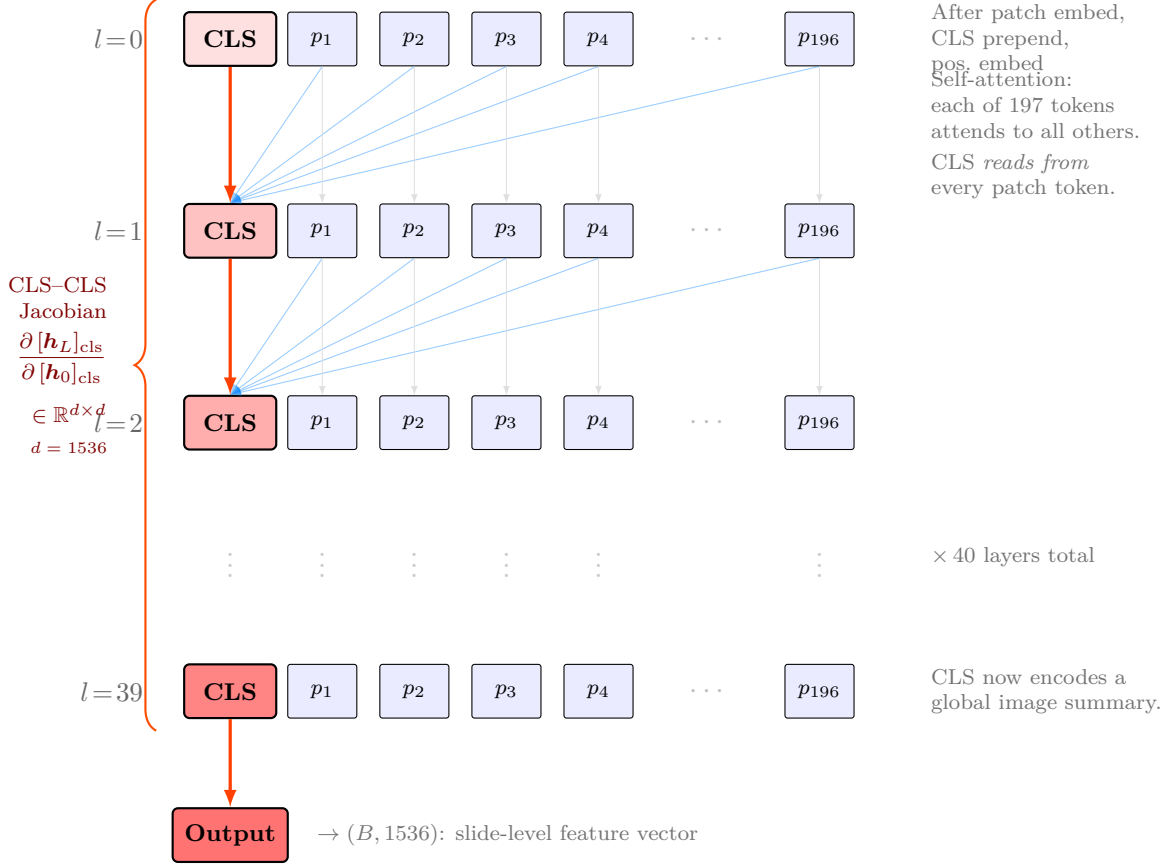


Figure 6: Information flow through the CLS token across transformer layers. At each layer, self-attention allows the CLS token (red, left column) to read from all 196 patch tokens (blue arrows), progressively accumulating a global image summary. Thick orange arrows show the residual (identity) pathway. The CLS/CLS Jacobian $\frac{\partial [\mathbf{h}_L]_{\text{cls}}}{\partial [\mathbf{h}_0]_{\text{cls}}} \in \mathbb{R}^{d \times d}$ (left brace) captures how perturbations at the input CLS token propagate through all $L = 40$ layers, including indirect effects mediated by attention to patch tokens: the CLS output at layer $l+1$ depends on all 197 tokens at layer l (including patches that were themselves influenced by the CLS input at earlier layers).

6.2 Two modes of operation

We implement two complementary modes:

Mode 1: CLS-token-only (materialisable). If we restrict to the CLS token (the first row of the token sequence), the Jacobian reduces to

$$\mathbf{J}_l^{\text{cls}} = \frac{\partial [\mathbf{h}_{l+1}]_{\text{cls}}}{\partial [\mathbf{h}_l]_{\text{cls}}} \in \mathbb{R}^{d \times d}, \quad (6)$$

where $d = 1536$. This 1536×1536 matrix can be materialised directly via `torch.autograd.functional.jacobian` and decomposed with standard dense SVD in ~ 1 second on GPU. This gives a complete spectrum

of all 1536 singular values.

Mode 2: Full-sequence matrix-free (via JVP/VJP). For the full $Nd \times Nd$ Jacobian, we avoid materialisation entirely. We define a `scipy.sparse.linalg.LinearOperator` that implements the matrix-vector products

$$\text{matvec: } \mathbf{u} = \mathbf{J} \mathbf{v} \quad \text{via} \quad \text{torch.func.jvp} \quad (7)$$

$$\text{rmatvec: } \mathbf{v} = \mathbf{J}^T \mathbf{u} \quad \text{via} \quad \text{torch.func.vjp} \quad (8)$$

and then call `scipy.sparse.linalg.svds` to obtain the top- k and bottom- k singular values via implicitly-restarted Lanczos bidiagonalisation. Each call to (7) or (8) has the cost of one forward or backward pass through a *single* transformer block.

6.3 Overall cost

Mode	Cost per layer	Output
CLS-only	$\sim d^2$ forward ops + dense SVD	All d singular values
Matrix-free, top/bottom k	$\sim (2k \cdot n_{\text{iter}})$ block fwd/bwd passes	Top- k and bottom- k singular values

Probing 5 evenly-spaced blocks with $k = 10$ in matrix-free mode requires $\sim 5 \times 2 \times 10 \times 20 = 2000$ block evaluations (assuming $n_{\text{iter}} \approx 20$ Lanczos steps), versus 40 blocks for one full forward pass. This is $\sim 50\times$ overhead *per probed sample*, so the probe should be run on a small batch (e.g. 1–4 samples) and not on every evaluation step. The CLS-only mode is far cheaper ($\sim 3\text{--}5\times$ one forward pass for 5 blocks) and is recommended for routine monitoring.

7 Implementation

7.1 SpectralProbe: standalone spectral estimator

Given a transformer block F_l and an activation tensor $\mathbf{h}_l$, the `SpectralProbe` estimates:

- **CLS mode:** materialises $\mathbf{J}_l^{\text{cls}} \in \mathbb{R}^{d \times d}$ via `torch.autograd.functional.jacobian`, then calls `torch.linalg.svdvals`.
- **Full mode:** wraps JVP/VJP in a `LinearOperator` and calls `scipy.sparse.linalg.svds` for top/bottom k .

7.2 SpectralBackboneEvaluator: integrated evaluator

`SpectralBackboneEvaluator` subclasses `SidebandBackboneEvaluator`. It reuses the existing hook infrastructure to capture intermediate activations at the probed blocks, then runs `SpectralProbe` on those activations after the forward pass. Results are stored in the `spectral_results` property, keyed by block index.

The evaluator is instantiated via `gigapath_backbone_evaluator` with name "GIGAPATH_SPECTRAL_BACKBONE_

7.3 Pipeline entry point

In `pipelines.py`:

```
gigapath_backbone_evaluator(  
    "GIGAPATH_BASELINE_BACKBONE_SPECTRAL_EVALUATOR",  
    device='cuda',  
)
```

This returns a `SpectralBackboneEvaluator` that probes 5 evenly-spaced blocks (indices 0, 10, 20, 30, 39) and estimates both the CLS-only full spectrum and the top/bottom 10 matrix-free singular values.

8 Interpretation Guide

Plot $\log \sigma_i^{(l)}$ versus layer index l for the top and bottom singular values. Expected patterns for a well-trained ViT:

1. **Most singular values near 1** (i.e. $\log \sigma \approx 0$): the network is approximately isometric, preserving the geometry of the representation.
2. **A few large singular values** ($\sigma \gg 1$): these correspond to directions the network selectively amplifies — likely the most semantically discriminative feature directions.
3. **A few small singular values** ($\sigma \ll 1$): these correspond to directions being aggressively compressed — the network is performing dimensionality reduction along these directions.
4. **Singular values converging as l increases**: this supports the attractor hypothesis — later layers are “closer” to the fixed point, with the Jacobian becoming more isometric.
5. **Condition number $\kappa^{(l)}$ decreasing with depth**: further evidence of convergence to an attractor with decreasing anisotropy.